

Predictive Modeling of Cardiovascular Disease Using Machine Learning

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Abstract

This research develops and compares two machine learning models—Logistic Regression and Random Forest—to predict cardiovascular disease (CVD) risk using a dataset of clinical and lifestyle factors. Through exploratory data analysis (EDA) on 68,731 patient records, key predictors such as systolic blood pressure, age, and cholesterol were identified. The Random Forest classifier demonstrated superior performance, achieving approximately 90% accuracy compared to Logistic Regression's 72%. The study utilizes SHAP (SHapley Additive exPlanations) to interpret the Random Forest model, providing clear insights into the factors driving its predictions. The findings highlight the effectiveness of ensemble methods for CVD risk assessment and the importance of model interpretability in clinical applications.

Keywords

Cardiovascular Disease, Machine Learning, Predictive Modeling, Random Forest, Logistic Regression, SHAP

1. Introduction

Cardiovascular diseases (CVDs) encompass a range of conditions that affect the heart and blood vessels, representing a significant global health burden. The early identification of individuals at high risk for CVDs is crucial for implementing preventive measures and reducing mortality rates. Traditional risk assessment methods often rely on limited factors and may not capture the complex interplay of various clinical and lifestyle variables.

This project addresses the challenge of accurate CVD risk prediction by leveraging machine learning techniques. The primary objective is to compare a linear model (Logistic Regression) with an ensemble model (Random Forest) to determine which can more effectively classify individuals as either healthy or at risk of cardiovascular disease. The scope of this research includes a thorough exploratory data analysis (EDA), model implementation and evaluation, and a SHAP-based interpretation of the top-performing model to provide actionable insights.

2. Literature Review

The application of machine learning in healthcare for disease prediction has been a rapidly growing field of research. Numerous studies have explored the use of various algorithms for predicting CVDs, with notable success. Common risk factors identified in the literature include age, gender, blood pressure, cholesterol levels, and lifestyle choices such as smoking and alcohol consumption.

Previous work in this area has compared the effectiveness of various models, from traditional statistical methods like Logistic Regression to more complex ensemble methods like Random Forest. While Logistic Regression provides a highly interpretable baseline, studies often show that ensemble methods like Random Forest can capture more complex, non-linear relationships within the data, frequently leading to higher predictive accuracy. This project contributes to this body of work by performing a direct comparison and utilizing modern interpretability techniques like SHAP to explain the "black box" nature of the more complex model.

3. Methodology

3.1. Research Design

This study employs a quantitative research design to develop and evaluate predictive models for cardiovascular disease. The workflow includes data cleaning, exploratory data analysis, and feature engineering. The dataset was subsequently split into training (80%) and testing (20%) sets, and features were scaled before training and comparing two models: a Logistic Regression model and a Random Forest classifier.

3.2. Data Collection

The dataset used in this project is a comprehensive collection of patient data, originally containing 70,000 records and 13 features. The features include demographic information (age, gender), clinical measurements (height, weight, systolic and diastolic blood pressure, cholesterol, glucose), and lifestyle factors (smoking, alcohol intake, physical activity). After data cleaning and preprocessing, the final dataset used for analysis consisted of 68,731 records.

3.3. Tools and Techniques

The project was implemented using the Python programming language with the following libraries:

- **Pandas & NumPy:** For data manipulation and numerical operations.
- **Matplotlib & Seaborn:** For data visualization.
- **Scikit-learn:** For implementing the Logistic Regression and Random Forest models.
- **SHAP (SHapley Additive exPlanations):** For interpreting the Random Forest model's

predictions.

3.4. Data Preprocessing

The initial dataset underwent several preprocessing steps to ensure data quality and suitability for modeling:

- **Age Conversion:** The 'age' column, originally in days, was converted to years and rounded to the nearest integer.
- **Blood Pressure Cleaning:** Anomalous blood pressure readings were removed by filtering out records based on statistical percentiles and logical checks (e.g., diastolic pressure cannot be higher than systolic).
- **Categorical Data Encoding:** Categorical features were converted into a numerical format through label encoding.
- **Duplicate Removal:** The dataset was checked for duplicate entries, and none were found.

4. Results

4.1. Exploratory Data Analysis (EDA)

The EDA revealed significant correlations between several features and the presence of cardiovascular disease. Systolic blood pressure (ap_hi), age, and cholesterol levels exhibit the strongest positive correlations with the target variable, cardio.

4.2. Machine Learning Model Performance

The two selected machine learning models were trained and evaluated on the preprocessed dataset. The Random Forest classifier demonstrated significantly better performance across all metrics. The results are summarized below:

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.90	0.88	0.84	0.86
Logistic Regression	0.72	0.74	0.70	0.72

4.3. Feature Importance with SHAP

Using SHAP (SHapley Additive exPlanations) on the superior Random Forest model allowed for a detailed interpretation of its predictions. The SHAP summary plot quantified the impact of each feature, confirming the findings from the EDA. The most significant predictors were:

1. **Systolic Blood Pressure (ap_hi)**
2. **Age**
3. **Cholesterol**
4. **Weight**

This analysis confirms that higher values for these features consistently push the model's prediction toward a higher risk of cardiovascular disease.

5. Discussion

The results clearly indicate the superiority of the Random Forest model over Logistic Regression for this specific prediction task. The 18-point difference in accuracy highlights the presence of complex, non-linear relationships in the data that the linear model could not capture.

The use of SHAP provides crucial transparency for the more complex Random Forest model. By confirming that the model's predictions are driven by clinically recognized risk factors (high blood pressure, age, cholesterol), we can be more confident in its utility. This addresses a common criticism of using "black box" models in healthcare, demonstrating that their predictions can be explained in an intuitive way. The findings align with established medical knowledge and validate the model's clinical relevance.

6. Conclusion

This research successfully compared a linear and an ensemble machine learning model for predicting cardiovascular disease, with the Random Forest classifier achieving 90% accuracy. The project demonstrated the value of using advanced techniques like SHAP to ensure model interpretability. The Random Forest classifier, explained through SHAP, stands out as a promising and trustworthy tool for early CVD risk detection.

While the model shows high accuracy, it is limited by the scope of the available dataset. Future work could involve incorporating additional features to potentially improve the model's predictive power further. Deploying the model in a real-world clinical setting for validation would be a crucial next step toward its practical application in healthcare.

7. References

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